

Chapter 3: Scatterplots and Correlation

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Part I: Exploring Data

Bivariate Data

- In Chapters 1 and 2, we examined one variable at a time. For example, we might study the distribution of test scores or household incomes. This is called **univariate data analysis**.
- Many statistical questions involve studying how two variables are related.
- For example, we might ask whether study time is related to exam score or whether temperature is related to electricity use.
- When two quantitative variables are measured on the same individual or observational unit, we have **bivariate data**.

Explanatory vs. Response Variables

When studying the relationship between two variables, it is useful to identify the role of each variable.

- **Response Variable (y):** The outcome variable that we want to explain or predict.
- **Explanatory Variable (x):** The variable that may help explain or predict changes in the response variable.

Note on Terminology

The explanatory variable is sometimes called the *predictor* or *independent variable*. The response variable is sometimes called the *dependent variable* or *target variable*.

Assigning Variables

Ex 3.1: Enzyme Activity and Temperature

Students run an enzyme reaction at different water bath temperatures and measure how much oxygen gas is produced.

Variable	Measurement	Biological Role
Explanatory (x)	Temperature ($^{\circ}\text{C}$)	Students manipulate this to drive the reaction speed.
Response (y)	Oxygen Gas (mL)	The biological outcome being measured.

Ex 3.2: Forest Ecology and Plant Growth

An ecologist measures the percentage of open sky (canopy cover) and the height of the tallest fern in 50 forest plots.

Variable	Measurement	Biological Role
Explanatory (x)	Canopy Open Sky (%)	The physical environment is the underlying cause.
Response (y)	Fern Height (cm)	The biological growth response.

Ex 3.3: Beer and Blood Alcohol

Student volunteers drank different amounts of beer and a police officer measured their blood alcohol concentration (BAC).

- Explanatory (x): Number of beers consumed
- Response (y): Percent alcohol in blood

Ex 3.4: Height

Statistics Canada's Canadian Health Measures Survey (CHMS) collects health information and physical measurements from Canadians, including height, age and sex. When the goal is simply to describe these measurements, there may be no explanatory or response variable.

However, if a researcher uses the data to study how height changes with age, then Age would be the **Explanatory** variable (x) and Height would be the **Response** variable (y).

Scatterplots

Once we have identified our explanatory (x) and response (y) variables, the most effective way to display their relationship is a **scatterplot**.

Rules for Building a Scatterplot:

- 1 Plot the **Explanatory Variable on the horizontal (x) axis**.
- 2 Plot the **Response Variable on the vertical (y) axis**.
- 3 Each individual appears as exactly **one dot** on the graph, located at the intersection of their x and y values.

Examining a Scatterplot

What to look for

In any graph of data, look for the overall pattern and for striking deviations from that pattern.

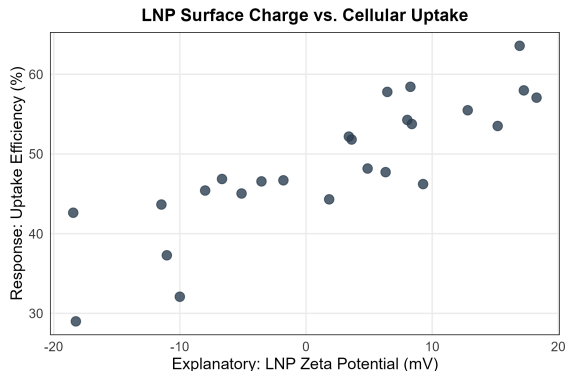
We describe the overall pattern of a scatterplot using four characteristics:

- **Form:** Is the relationship *Linear* (straight line) or *Non-Linear* (curved, U-shaped)?
- **Direction:** *Positive association* (slopes upward) or *Negative association* (slopes downward).
- **Strength:** How tightly do the points cluster around the underlying form? (*Strong, moderate or weak*).
- **Outliers:** Are there points that fall far outside the overall cloud?

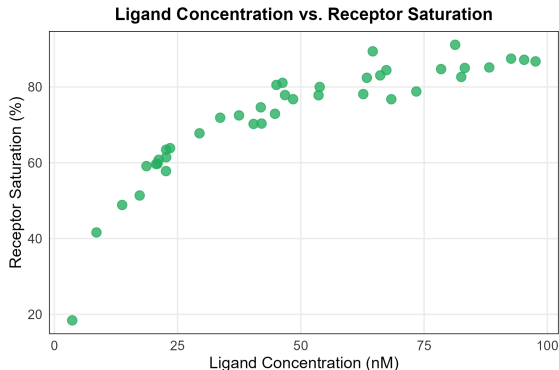
Example 3.5: Linear Relationships

Context: Researchers are studying whether the surface charge of a Lipid Nanoparticle (LNP) is related to how efficiently it enters cells. They prepare several LNP formulations with different surface charges and measure the percentage of particles taken up by cells. Each point represents one LNP formulation.

- **Explanatory variable (x):** LNP surface charge
- **Response variable (y):** Cellular uptake efficiency
- **Form:** Linear
- **Direction:** Positive
- **Strength:** Moderate-to-Strong
- **Outliers:** None apparent



Example 3.6: Non-Linear Relationships



Context: Researchers expose cells to different concentrations of a molecule that binds to receptors on the cell surface. For each concentration, they measure the percentage of receptors that become occupied. Each point represents one concentration and its corresponding receptor saturation.

- **Explanatory variable (x):** Ligand concentration
- **Response variable (y):** Receptor saturation
- **Form:** Strong and non-linear

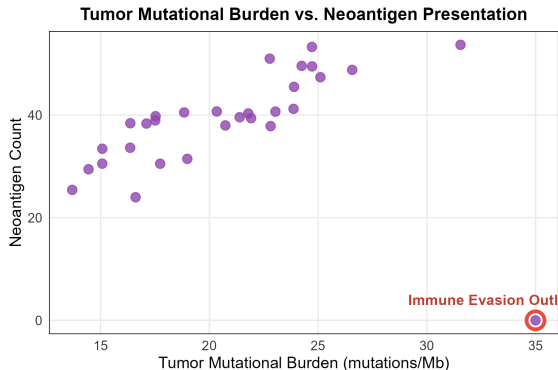
Why does the curve flatten?

At low concentrations, increasing the ligand concentration produces a large increase in receptor saturation. As more receptors become occupied fewer remain available. The response therefore begins to level off as saturation approaches its upper limit.

Example 3.7: Outliers

Context: Researchers are studying whether tumours with more genetic mutations also produce more neoantigens. Neoantigens are abnormal proteins created by tumour mutations that may be recognized by the immune system. Each point represents one tumour sample.

- **Explanatory variable (x):** Tumour Mutational Burden (TMB), which measures the number of mutations in the tumour
- **Response variable (y):** Number of neoantigens presented by the tumour
- **Overall Pattern:** Positive, approximately linear and moderately strong
- **Outlier:** One tumour near (35, 0) has a high TMB but almost no observed neoantigen presentation

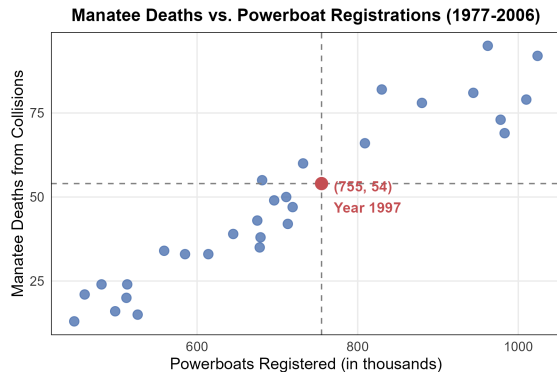


Possible explanation: A tumour may contain many mutations but still fail to present the resulting neoantigens effectively. For example, changes affecting the HLA antigen-presentation system could reduce what is displayed to the immune system.

Positive vs. Negative Association

- Two variables are **positively associated** when above average values of one variable tend to accompany above average values of the other.
- Two variables are **negatively associated** when above average values of one variable tend to accompany below average values of the other.

Example 3.8: Manatees and Powerboats



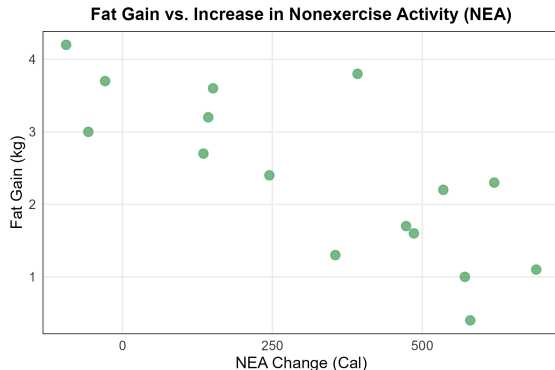
Context: Researchers are studying whether increased boating activity is associated with the number of manatee deaths caused by collisions with watercraft in Florida. Each point represents one year of observations.

- **Explanatory variable (x):** Number of registered powerboats in Florida
- **Response variable (y):** Number of manatee deaths caused by watercraft collisions
- **Direction:** Positive. Years with more registered powerboats tend to have more manatee deaths.
- **Form:** Approximately linear
- **Strength:** Strong. The points follow the overall increasing pattern fairly closely.

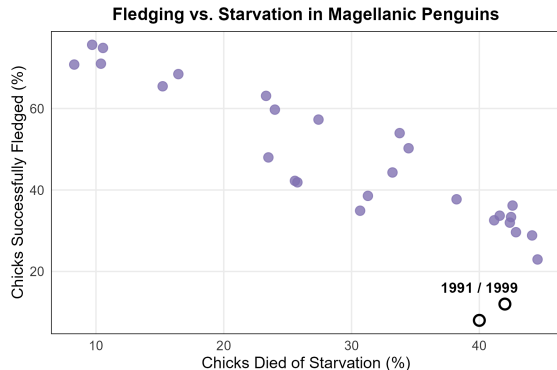
Example 3.9: Does Fidgeting Keep You Slim?

Context: Researchers deliberately increased the food intake of a group of volunteers and measured how much body fat each person gained. They also measured changes in **Nonexercise Activity (NEA)**, which includes everyday movement such as walking, standing and fidgeting rather than planned exercise. Each point represents one volunteer.

- **Explanatory variable (x):** Change in Nonexercise Activity (NEA)
- **Response variable (y):** Fat gain in kilograms
- **Direction:** Negative. Volunteers with larger increases in NEA tended to gain less fat.
- **Form:** Approximately linear
- **Strength:** Moderately strong. The points follow a decreasing pattern but show noticeable variation around that pattern.



Example 3.10: Magellanic Penguins



Context: Researchers followed a colony of Magellanic penguins over many breeding seasons. For each year, they recorded the percentage of chicks that died from starvation and the percentage that successfully fledged, meaning that they survived long enough to leave the nest. Each point represents one breeding year.

- **Explanatory variable (x):** Percentage of chicks that died from starvation
- **Response variable (y):** Percentage of chicks that successfully fledged
- **Direction:** Negative. Years with more starvation generally had fewer chicks successfully fledging.
- **Form:** Approximately linear
- **Outliers:** The years 1991 and 1999 fall well below the overall pattern.

Correlation (r)

Scatterplots rely on subjective visual judgment. To make reproducible claims, we need a mathematical summary.

Pearson's Correlation Coefficient (r) measures the direction and exact strength of the *linear* relationship between two quantitative variables.

The formula standardizes both variables (converting them into z-scores), multiplies them together and averages them:

$$r = \frac{1}{n-1} \sum \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right)$$

Crucial Properties of r

- 1 r is always a number strictly between -1 and $+1$.
- 2 $r > 0$ indicates a positive linear association; $r < 0$ indicates a negative linear association.
- 3 Values near 0 indicate a very weak linear relationship. Values near -1 or $+1$ indicate points that lie nearly exactly on a straight line.
- 4 r **only measures linear strength**. It is completely blind to curved/non-linear relationships.

Example 3.11: The Mechanics of Correlation (Frog Locomotion)

A biologist measures the snout-vent length (cm) and maximum leap distance (cm) of 5 juvenile bullfrogs.

Frog ID	Length x (cm)	Leap Distance y (cm)
1	4.0	32.0
2	5.0	42.0
3	6.0	48.0
4	7.0	55.0
5	8.0	65.0

Question: What is Pearson's correlation coefficient between frog length and leap distance?

Example 3.11: Calculating Pearson Correlation

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Step 1: Calculate the means and standard deviations

$$\begin{aligned}\bar{x} &= 6.0 & \bar{y} &= 48.4 \\ s_x &\approx 1.581 & s_y &\approx 12.542\end{aligned}$$

Step 2: Standardize each observation

x	y	$z_x = \frac{x - \bar{x}}{s_x}$	$z_y = \frac{y - \bar{y}}{s_y}$	$z_x z_y$
4	32	-1.265	-1.308	1.654
5	42	-0.632	-0.510	0.323
6	48	0.000	-0.032	0.000
7	55	0.632	0.526	0.333
8	65	1.265	1.324	1.674
Sum				3.984

Step 3: Substitute into the correlation formula

$$r = \frac{1}{n-1} \sum z_x z_y = \frac{3.984}{5-1} \approx 0.996$$

The correlation is approximately $r = 0.996$. The data show a very strong positive linear association: longer juvenile frogs tend to have greater leap distances.

Example 3.11: Correlation Using Sums

This equivalent formula calculates correlation directly from the deviations from the two means:

$$r_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

For the frog data

$$\bar{x} = 6.0 \quad \text{and} \quad \bar{y} = 48.4$$

x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})^2$	$(y_i - \bar{y})^2$	$(x_i - \bar{x})(y_i - \bar{y})$
4	32	-2	-16.4	4	268.96	32.8
5	42	-1	-6.4	1	40.96	6.4
6	48	0	-0.4	0	0.16	0.0
7	55	1	6.6	1	43.56	6.6
8	65	2	16.6	4	275.56	33.2
Sum		0	0	10	629.20	79.0

$$r = \frac{79.0}{\sqrt{(10)(629.20)}} = \frac{79.0}{79.322} \approx 0.996$$

Example 3.11: Correlation in R (Method 1)

Method 1: Using the Formula

```
length_x <- c(4, 5, 6, 7, 8)
leap_y <- c(32, 42, 48, 55, 65)

# Store the number of observations
n <- length(length_x)

# Convert the x-values to z-scores
z_x <- (length_x - mean(length_x)) / sd(length_x)

# Convert the y-values to z-scores.
z_y <- (leap_y - mean(leap_y)) / sd(leap_y)

# Calculate Pearson correlation coefficient
r_manual <- sum(z_x * z_y) / (n - 1)

r_manual

[1] 0.995939
```

Example 3.11: Correlation in R Using Sums

Using the sum-of-deviations formula

```
length_x <- c(4, 5, 6, 7, 8)
leap_y <- c(32, 42, 48, 55, 65)

# Calculate deviations from the means
x_deviation <- length_x - mean(length_x)
y_deviation <- leap_y - mean(leap_y)

# Calculate the numerator
numerator <- sum(x_deviation * y_deviation)

# Calculate the two sums in the denominator
sum_x_squared <- sum(x_deviation^2)
sum_y_squared <- sum(y_deviation^2)

# Calculate Pearson's correlation coefficient
denominator <- sqrt(sum_x_squared * sum_y_squared)
r_sums <- numerator / denominator

r_sums
[1] 0.995939
```

Example 3.11: Correlation in R (Method 2)

Method 2: Creating a Custom Function

```
calculate_r <- function(x, y) {  
  
  # Store the number of paired observations  
  n <- length(x)  
  
  # Standardize both variables.  
  z_x <- (x - mean(x)) / sd(x)  
  z_y <- (y - mean(y)) / sd(y)  
  
  # Calculate and return Pearson's correlation  
  return(sum(z_x * z_y) / (n - 1))  
}  
  
calculate_r(length_x, leap_y)  
  
[1] 0.995939
```

Example 3.11: Correlation in R (Method 3)

While it's important to understand the math, R has a built-in shortcut for calculating Pearson's correlation coefficient.

Method 3: The fast way

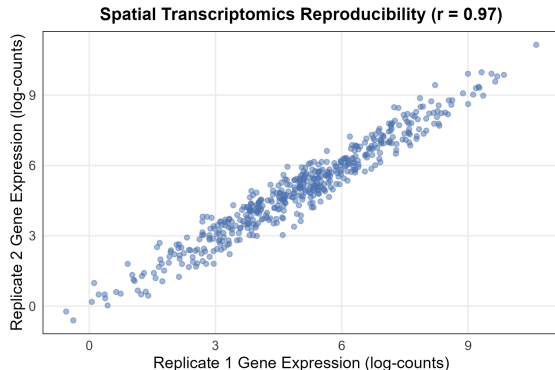
```
# In practice, you will simply use the cor() function:  
cor(length_x, leap_y)  
[1] 0.995939
```

Interpretation: An r of roughly 0.996 tells us there is a nearly perfect, positive linear relationship between a juvenile frog's length and how far it can leap.

Example 3.12: Calculating r (Spatial Reproducibility)

Context: A research lab uses a spatial transcriptomics assay to measure gene activity in a tissue sample. To check whether the laboratory procedure gives consistent results, the same type of experiment is performed twice under similar conditions. For each of 500 genes, the researchers compare the measured expression level from Replicate 1 with the corresponding expression level from Replicate 2. Each point represents one gene.

- **Explanatory variable (x):** Gene expression measured in Replicate 1
- **Response variable (y):** Gene expression measured in Replicate 2
- **Correlation:** $r = 0.9696$
- **Direction:** Positive
- **Strength:** Very strong linear association



Interpretation: Genes with higher expression in the first replicate also tend to have higher expression in the second replicate. The large value of r suggests strong reproducibility of the relative gene-expression pattern.

Example 3.13: The Unit-Independence of r

Because the formula divides by standard deviations (s_x and s_y), it entirely strips away the physical units of measurement.

Rule

Correlation has no units and changing the units of measurement does not change r .

If you calculate correlation between swimming speed (**km/h**) and energy expenditure (**Kcal**), you get $r = 0.9424$.

If a colleague converts the data to meters per second (**m/s**) and Joules (**J**), the correlation remains **identically 0.9424**.

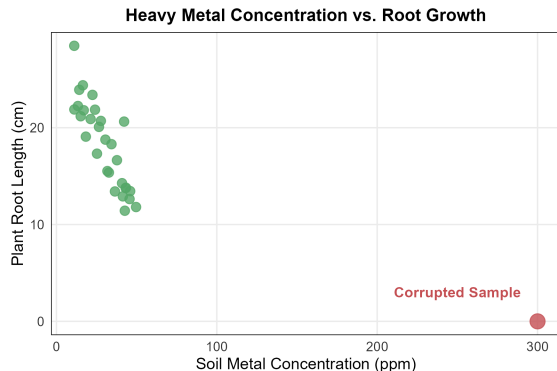
Example 3.14: The Non-Resistance of Correlation

Remember that the mean ($\bar{x}$) and standard deviation (s) are non-resistant to outliers.

Because the formula for r relies entirely on $\bar{x}$ and s , **correlation is strongly non-resistant to outliers.**

A single extreme point can completely destroy a strong correlation or artificially create one out of thin air! Let's look at a soil toxicology study.

Example 3.14: Heavy Metal and Root Length



One unusually extreme industrial sample was then included. Its metal concentration was far outside the range of the other observations and its root length was very small. After adding this one point, the correlation changes to approximately $r = -0.95$.

Context: Researchers are studying whether higher concentrations of a heavy metal in soil are associated with reduced plant root growth. For each soil sample, they record the heavy metal concentration and the root length of a plant grown in that soil. Each point represents one soil sample.

- **Explanatory variable (x):** Heavy metal concentration in the soil
- **Response variable (y):** Plant root length
- **Original pattern:** The 29 regular samples show a moderately strong negative linear association with $r = -0.77$

Summary of R Commands (Chapter 3)

Here are the key functions introduced in this chapter for bivariate analysis:

R Command	What it Calculates
<code>length(x)</code>	Returns the number of elements (n) in a vector.
<code>sum(x * y)</code>	Multiplies two vectors together element-by-element, then sums the result.
<code>function(x, y) {...}</code>	Allows you to create your own custom, reusable mathematical formulas.
<code>cor(x, y)</code>	Calculates Pearson's Correlation Coefficient (r) between two variables instantly.